

A Taxonomy of Construction Task Activities for Robot Workers

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Abstract—Recent vision–language–action models offer a path toward robots with broader repertoires than conventional task-specific systems. Construction deployment, however, requires a precise inventory of worker activities and the capabilities needed to execute them. We present TARGAT, an occupation-grounded taxonomy derived from 91 O*NET tasks across seven high-employment construction occupations and 30 instructional videos of physical work. TARGAT defines 41 action primitives in 12 groups and three classes and provides a mechanism for composing parameterized primitive sequences into reusable skills. This human-interpretable structure can organize demonstrations, specify robot requirements, and support coding agents that retrieve and extend skill libraries. We also demonstrate selected primitives on a DOBOT CR3 arm with a CRAFT hand. TARGAT thereby provides a common vocabulary for analyzing human work and developing general-purpose construction robots. Annotations are available at <https://github.com/AICPS/TARGAT-Taxonomy>.

I. INTRODUCTION

Vision–language–action (VLA) models are broadening robot repertoires across embodiments, scenes, and instructions [1]. Construction robotics likewise spans earthmoving, additive manufacturing, assembly, and finishing, yet deployed systems remain task-specific because materials, tools, and sites vary substantially [2]. Generalist construction robots therefore require an explicit specification of primitive capabilities and their composition into complex work. Such a specification is needed both to teach robots what to learn and to evaluate whether their repertoires cover the activities performed by human workers.

Action taxonomies provide this interface between occupational work and robot learning. TARMAC, for example, decomposes chemistry procedures into reusable manipulation actions [3], but does not cover construction-specific tool use, whole-body motion, communication, or information processing. Existing construction classifications address field operations or human–robot interaction rather than occupation-grounded, composable actions. Consequently, they cannot directly identify capability gaps or describe how reusable robot behaviors form a construction task. TARGAT (Taxonomy of Construction Task Activities for Robot Workers) fills this gap.

Robot coding agents such as ASPIRE generate and validate control programs, retain successful procedures in skill libraries, and retrieve them for new tasks [4]. TARGAT can

ground this process: its primitives provide a stable action vocabulary, parameterized skills encode compositions, and task decompositions provide examples that an agent can retrieve, adapt, and extend.

TARGAT contributes (1) 41 action primitives in 12 groups and three classes; (2) a mechanism to create reusable skills as ordered, parameterized primitive sequences; (3) annotations of 30 videos covering 91 O*NET tasks across seven occupations; and (4) an initial physical instantiation on a collaborative arm with a dexterous hand. Together, these elements provide a common vocabulary for demonstrations, robot requirements, and skill composition across embodiments (Fig. 1). TARGAT is therefore a capability vocabulary, not a fixed control architecture: it specifies what a robot must accomplish without prescribing a particular implementation. The same representation can index human demonstrations, structure robot skill libraries, and expose missing capabilities across embodiments.

II. RELATED WORK

Construction robotics and task scope. Construction robots perform earthmoving, material deposition, assembly, installation, and finishing, but most remain specialized and require structured sites or substantial adaptation [2]. Broader platforms have learned eight humanoid construction actions from demonstrations [5] or supported teleoperated transport, screw-gun use, and overhead inspection [6]. These studies establish the feasibility of individual capabilities but do not define the action space of a generalist construction robot. At the process level, Everett and Slocum organized field operations into seven levels and proposed 12 basic automation tasks [7], while Rodrigues et al. classify human–robot interaction by platform, autonomy, interaction, and team structure [8]. TARGAT instead provides an occupation-grounded vocabulary of worker actions and their compositions, complementing these system- and process-level views.

Human-action and manipulation representations. A proposed manipulation vocabulary include 32 cooking motions characterized by contact, trajectory, duration, and manual-operation attributes [9], while TARMAC derives robot-executable primitives from chemistry practice [3]. Recognition resources operate at a different level: AVA annotates 80 atomic actions in movies [10], and the Construction Motion Library provides 61,275 construction-related 3-D skeleton sequences [11]. These resources support manipulation or recognition but do not represent construction workflows spanning physical, cognitive, and communicative actions. Manipulation taxonomies omit whole-body process structure,

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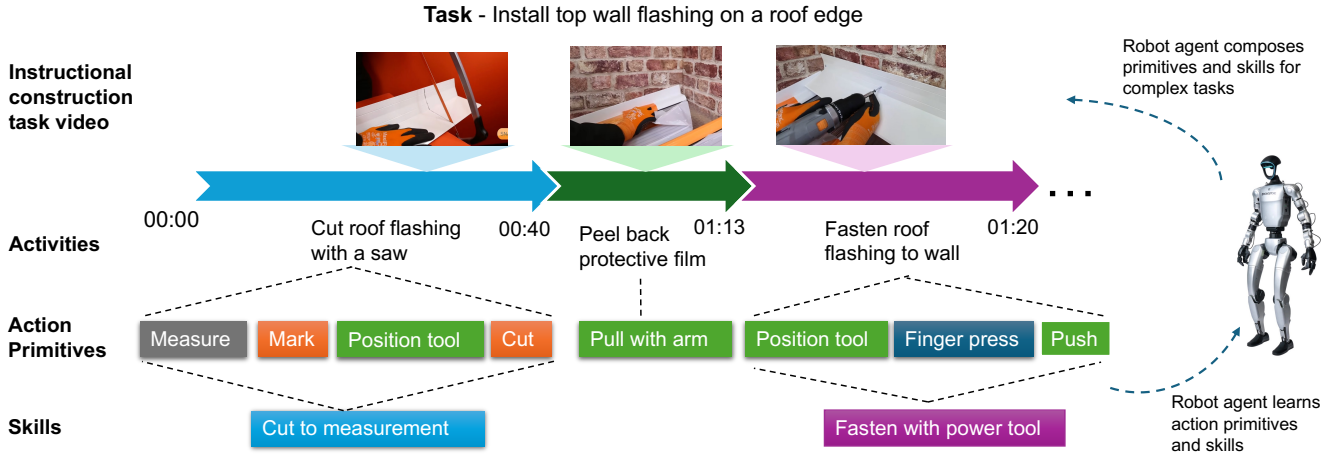


Fig. 1. **Overview of TARCAT.** Instructional construction videos are segmented into observable activities, labeled with reusable action primitives, and organized into skills. Robot agents can learn, retrieve, and compose these elements to execute complex tasks.

general video labels omit trade context and ordered workflows, and motion datasets do not specify robot-executable tool and contact skills. TARCAT bridges these levels by combining fine-grained primitives with compositional skills grounded in construction work.

Occupational grounding. GDPval used BLS employment data and O*NET task statements to select economically relevant knowledge work [12]. TARCAT applies the same grounding to construction: BLS defines the occupational scope, and O*NET captures the physical and non-movement work within it. This grounding connects the taxonomy to economically significant occupations rather than a manually selected set of robot demonstrations.

III. INSTRUCTIONAL VIDEO COLLECTION

We defined the corpus using two U.S. Department of Labor resources. O*NET provides SOC-aligned occupational task statements [13], while the BLS Occupational Outlook Handbook reports 2024 employment for Construction and Extraction profiles and maps them to O*NET occupations [14]. Excluding heavy machinery operators, we selected the seven highest-employment profiles and mapped each to one O*NET SOC occupation: (i) Construction Laborers, (ii) Carpenters, (iii) Electricians, (iv) Plumbers, Pipefitters, and Steamfitters, (v) Painters, Construction and Maintenance, (vi) Brickmasons and Blockmasons, and (vii) Roofers.

These occupations comprise 169 O*NET task statements, of which 137 require physical movement and 32 do not. Using ChatGPT with GPT-5.6 [15], we searched YouTube for instructional demonstrations of the movement tasks. We retained videos with clearly observable, task-relevant actions and mapped each video to every task it demonstrates. The 30 videos cover 91 O*NET tasks across the selected occupations.

IV. ACTIVITY TAXONOMY

We analyzed O*NET task descriptions and instructional videos to derive context-independent actions that require distinct robot capabilities and compose into complex tasks. We term these *action primitives*. They serve as building

blocks that a robot can learn and compose into more complex behavior. Ordered primitive sequences that achieve a subgoal form reusable *skills* across tasks and occupations. The following subsections define the primitive hierarchy and illustrate skill and task decomposition.

We distinguish primitives by the capability exercised rather than by occupation or tool identity. Activities performed with different tools share a primitive when they impose the same motion or interaction requirement; superficially similar activities remain distinct when they require different contact, force, or coordination patterns. This abstraction supports transfer across trades while preserving control-relevant distinctions.

A. Primitives

For non-movement tasks, we inferred activities from O*NET statements. For movement tasks, a human annotated timestamped activities in each demonstration. Each activity received a primitive label; sequences that achieved a subgoal received a skill label. This procedure preserves the relationship between an occupational task, its observable execution, and the reusable capabilities it requires.

TARCAT contains 41 primitives in three classes: information processing, communication, and tool use. *Classes* distinguish broad roles, while *groups* collect primitives with a shared function or motion pattern. Table I presents this hierarchy. Here, *object* denotes either a tool or another manipulated item; *tool* is used when no separate object is involved. This convention allows each primitive to generalize across trades and specific equipment. The classes and groups are defined below.

- 1) **Information Processing:** Acquire, interpret, and use task-relevant information to determine subsequent activities.
 - **Assessment:** Assess relevant conditions to establish the state of the task or work environment.
 - **Comprehension:** Interpret information or signals to establish their intended meaning and requirements.

TABLE I
ACTION PRIMITIVES AND THEIR HIERARCHICAL ORGANIZATION.

Class	Group	Action Primitive	Examples
Information Processing	Assessment	Take measurements	Read pressure gauge; measure roof width
		Determine tool-application location	Locate drill point; locate nail-gun cap
		Inspect to check state or find issues	Inspect roof damage; check ladder locks
Information Processing	Comprehension	Study plan, instruction or specifications	Read compressor label; check breaker rating
		Listen to or observe instructions or signals	Follow spoken directions; observe crane signals
		Estimate time, materials, labor and cost	Estimate roofing labor; calculate mortar quantity
Information Processing	Planning	Select and procure tools and materials	Order lumber; choose paint finish
		Prepare design, layout or installation plans	Plan wiring layout; sketch equipment locations
Communication	Collaboration	Talk to convey information	Discuss work plan; explain safety issue
		Use gestures to signal other workers	Signal crane operator; direct team lift
		Schedule and coordinate work with others	Schedule work crew; arrange subcontractor
Communication	Documentation	Prepare records, documents, and reports	Write project report; record inspection result
Tool Use	Finger manipulation	Pull or push an object with fingers	Open nail-gun cap; connect air hose
		Press an object with fingers	Press laser button; squeeze nail-gun trigger
		Open fingers	Release plumb bob; release stud finder
		Rotate an object with fingertips	Adjust pressure valve; tighten spray-can straw
	Arm translation and positioning	Grab an object	Catch plumb-bob string; grab panel cover
		Pull or push an object with arm	Open cabinet door; push breaker into slot
		Pick or place an object	Pick up trowel; place electrical box
		Position an object	Align roof flashing; position screwdriver
	Arm and wrist rotation	Move tool along a linear path	Apply joint sealant; dip brush into paint
		Twist an object using wrist rotation	Open cement bottle; twist wire with pliers
		Pivot a tool about a fixed point	Tighten bolt with wrench; turn valve key
		Swing arm while holding an object	Hammer a nail; chip mortar with chisel
Tool Use	Mixing motion	Reorient an object	Flip shingle; turn brick over
		Pour liquid from a container	Pour paint into tray; pour water into bucket
		Scoop material with a tool	Scoop mortar with trowel; scoop soil with shovel
	Surface-contact manipulation	Shake an object	Shake spray can; shake mixing container
		Stir liquid and solid mixture	Mix mortar with trowel; stir paint with stick
		Wrap around, tie or untie	Tie rope around ladder; untie plumb-bob knot
	Whole-body locomotion	Mark on a surface	Mark stud line; mark screw point
		Cut an object	Saw wooden stud; slice roofing shingle
		Spread object on surface	Spread mortar; smooth joint compound
		Stroke tool on surface	Brush paint on wall; roll paint on ceiling
Tool Use	Whole-body load handling	Walk to a location	Walk to inspection area; approach electric outlet
		Carry an object to a location	Carry measuring rod; carry roll of felt
		Climb up or down	Climb ladder; descend scaffold
	Whole-body load handling	Pull or push a heavy object	Align wall cabinet; slide loaded pallet
		Pick or place a heavy object	Pick up ladder; place wall cabinet
		Carry a heavy object	Carry wall cabinet; transport ladder
		Bend down or straighten up	Pick up leaf blower from ground; lift drill and stand
	Whole-body load handling		

- **Planning:** Estimate resources, select tools, and develop designs or layouts before execution.
- 2) **Communication:** Exchange or preserve task-relevant information to coordinate work.
 - **Collaboration:** Convey information and coordinate activities with co-workers.
 - **Documentation:** Create written documents to communicate and preserve information.
- 3) **Tool Use:** Perform embodied motions that manipulate tools or objects and exert physical effort on the work environment.
 - **Finger manipulation:** Manipulate primarily through finger motion or force, with little gross arm movement.
 - **Arm translation and positioning:** Grab or move an object through coordinated hand and arm motion.
 - **Arm and wrist rotation:** Rotate or reorient the arm,

wrist, tool, or object to apply force.

- **Mixing motion:** Use repeated, circular, or wrapping motions to mix or secure materials.
- **Surface-contact manipulation:** Move a tool while maintaining controlled contact with a work surface.
- **Whole-body locomotion:** Use the legs and torso to move between locations or transport an object.
- **Whole-body load handling:** Use whole-body force and postural control to manipulate heavy objects or bend and straighten.

B. Skill and Task Decomposition

A skill is an ordered sequence of activities labeled with action primitives. Because skills are defined independently of task context, they may contain *variables* that are instantiated with the tools or objects observed in a task segment. Table II illustrates this representation for disconnecting a wire from a circuit breaker.

TABLE II
ACTION SEQUENCE FOR A WIRE-DISCONNECTION SKILL.

Skill	Disconnect wire by loosening terminal screw with a screwdriver	
Variables	terminal	
Steps	<i>Activity</i>	<i>Action primitive</i>
	Rotate the screwdriver to loosen terminal screw holding the wire	Twist an object using wrist rotation
	Pull the wire out of the terminal	Pull or push an object with arm
Subtask	Disconnect wire from a circuit breaker. terminal=circuit breaker terminal.	

TABLE III
DECOMPOSITION OF A TILE-ATTACHMENT TASK.

Video	Attach a tile vertically below a point		
	Activity	Action primitive	Skill
	Untie plumb-bob knot	Wrap around, tie or untie	
	Position plumb bob	Position an object	
	Release plumb bob	Open fingers	Mark vertical guide
Activities	Stop plumb bob	Grab an object	
	Scoop mortar	Scoop material with a tool	Apply and spread mixture
	Position trowel	Position an object	
	Apply mortar with trowel	Swing arm while holding an object	
	Spread mortar	Spread object on surface (repeated)	
	Position tile	Position an object	Attach object to a location
	Press tile onto wall	Pull or push an object with arm	

At the task level, skills and standalone primitives are composed into construction procedures. Three cases require explicit representation: simultaneous actions, repeated actions, and nested skills. We record simultaneous actions in an arbitrary sequential order, mark repetition with the repeated attribute, and allow a skill to invoke another skill as a subskill. Table III illustrates this decomposition for a tile-attachment demonstration.

V. REAL-WORLD DEPLOYMENT

We instantiated four tool-use activities on a six-axis DOBOT CR3 arm [16] with a 15-actuator, tendon-driven CRAFT hand [17]: positioning a power screwdriver, placing a hammer in a box, hammering a nail, and brushing objects aside. These scripted demonstrations span arm translation and positioning, arm and wrist rotation, and surface-contact manipulation. They confirm that selected taxonomy elements map to executable physical behaviors. A demonstration is available at <https://youtu.be/Dc-bcWGwTLc>.

VI. CONCLUSION

We presented TARCAT, an occupation-grounded representation of construction work based on 41 action primitives and a mechanism for composing them into parameterized skills. By linking O*NET tasks with instructional evidence, the taxonomy translates the scope of human construction work into a structured, human-interpretable robot capability specification. Its decompositions can also guide ASPIRE-like agents in mapping instructions to known primitives,

composing task-specific skills, and retaining validated compositions for reuse. The initial hardware study demonstrates the executability of selected primitives but remains preliminary. Other limitations include coverage of seven occupations and few demonstrations per task. Future work will expand the corpus, measure annotation agreement and coverage, implement more primitives on mobile manipulators and humanoids, and evaluate long-horizon skill composition.

REFERENCES

- [1] M. J. Kim *et al.*, “OpenVLA: An open-source vision-language-action model,” *arXiv preprint arXiv:2406.09246*, 2024.
- [2] N. Melenbrink, J. Werfel, and A. Menges, “On-site autonomous construction robots: Towards unsupervised building,” *Automation in Construction*, vol. 119, p. 103312, 2020.
- [3] K. Huang, J. Pipe, A. E. Martin, T. Wang, B. A. Franklin, A. M. Tyrrell, I. J. S. Fairlamb, and J. Zhu, “TARMAC: A taxonomy for robot manipulation in chemistry,” *arXiv preprint arXiv:2510.19289*, 2025.
- [4] R. Lu, Y. Wu, E. Kou, L. Fu, W. Xiao, A. Mandlekar, Y. Xu, G. Shi, K. Goldberg, A. Chen, M. Chowdhury, Y. Zhu, L. Fan, and G. Wang, “ASPIRE: Agentic /Skills Discovery for Robotics,” NVIDIA Technical Report, 2026. [Online]. Available: <https://research.nvidia.com/labs/gear/aspire/>
- [5] Y. Liu and Y. Liu, “Perception-and-action system for humanoid robot task execution in construction,” *Computer-Aided Civil and Infrastructure Engineering*, vol. 50, p. 100107, 2026.
- [6] X. Hu, Y. Shi, and Y. Ye, “Teleoperation of a semi-humanoid for construction with underactuated dexterous hands,” in *Computing in Civil Engineering 2025: Resilient, Robotic, and Educational Systems*. American Society of Civil Engineers, 2025, pp. 563–571.
- [7] J. G. Everett and A. H. Slocum, “Automation and robotics opportunities: Construction versus manufacturing,” *Journal of Construction Engineering and Management*, vol. 120, no. 2, pp. 443–452, 1994.
- [8] P. Rodrigues, R. Singh, M. Oytun, P. Adami, P. Woods, B. Becerik-Gerber, L. Soibelman, Y. Copur-Gencturk, and G. Lucas, “A multi-dimensional taxonomy for human–robot interaction in construction,” *Automation in Construction*, vol. 150, p. 104845, 2023.
- [9] D. Paulius, Y. Huang, J. Meloncon, and Y. Sun, “Manipulation motion taxonomy and coding for robots,” in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 5596–5601.
- [10] C. Gu, C. Sun, D. A. Ross, C. Vondrick, C. Pantofaru, Y. Li, S. Vijayanarasimhan, G. Toderici, S. Ricco, R. Sukthankar, C. Schmid, and J. Malik, “AVA: A video dataset of spatio-temporally localized atomic visual actions,” in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 6047–6056.
- [11] Y. Tian, H. Li, H. Cui, and J. Chen, “Construction motion data library: An integrated motion dataset for on-site activity recognition,” *Scientific Data*, vol. 9, no. 1, p. 726, 2022.
- [12] OpenAI, “GDPval: Evaluating AI model performance on real-world tasks,” OpenAI, Tech. Rep., 2025. [Online]. Available: <https://openai.com/index/gdpval/>
- [13] National Center for O*NET Development, “O*NET database,” 2026, accessed: 2026-08-20. [Online]. Available: <https://www.onetcenter.org/database.html>
- [14] U.S. Bureau of Labor Statistics, “Construction and extraction occupations,” *Occupational Outlook Handbook*, 2025, accessed: 2026-08-20. [Online]. Available: <https://www.bls.gov/ooh/construction-and-extraction/home.htm>
- [15] OpenAI, “GPT-5.6: Frontier intelligence that scales with your ambition,” Jul. 2026, accessed: 2026-08-23. [Online]. Available: <https://openai.com/index/gpt-5-6/>
- [16] Dobot, “DOBOT CR3 specifications,” 2026, accessed: 2026-08-20. [Online]. Available: <https://www.dobot.us/cr3-specs/>
- [17] L. Lin, S. Patel, J. Moon, S. Lazebnik, and U. Jain, “CRAFT: A tendon-driven hand with hybrid hard–soft compliance,” *arXiv preprint arXiv:2603.12120*, 2026.